

Money and Success in Football

DOES FINANCIAL STRENGTH PREDICT LEAGUE POSITION?

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Sports Economics Seminar

Introduction

The Billion-Euro Question

- The five major European leagues spent over **€12 billion** on transfers in 2023–24 alone.
- Fan debate: does buying expensive squads actually *buy* titles?
- Or can a **well-organised, underfunded club** still win?

This study uses 10 years of data from 5 leagues (975 team-seasons) to answer with evidence, not opinion.

Three Research Questions

1. Does financial rank predict league position?
2. *How much* of the variation does money explain?
3. Who defies the financial logic entirely?

Data & Methodology

Five Leagues, Ten Seasons – 975 Observations

League	N	Seasons	$r(\text{FR}, \text{Pos})$
Bundesliga	180	10	0.783
La Liga	200	10	0.787
Ligue 1	196	10	0.791
Premier League	200	10	0.801
Serie A	199	10	0.889
All	975	10	0.813

Seasons: 2015–16 to 2024–25. All r values significant at

$p < 0.001$.

Key Variable: Financial Rank (FR)

- Squads ranked by squad market value **within each league-season**.
- **FR = 1** = richest squad that season.
FR = 20 = least wealthy.
- Normalising within leagues removes between-league scale differences (Premier League values $\approx 3\times$ Bundesliga).

Statistical view: Spearman $\rho = 0.814$, Pearson $r = 0.813$; both $p < 0.001$.

Plain English: The wealth ranking of a squad is an almost perfect mirror of its final position.

Statistical Evidence

Richer Squads Finish Higher – Always

Richer Squads Finish Higher — With Notable Exceptions

N = 975 team-seasons • Pearson $r = 0.81$, $p < 0.001$ • Model H quadratic fit • Gold = key story clubs



Financial Rank normalised within each league-season. Axis reversed: top = better.

$$r = 0.813$$

Pearson correlation, Financial Rank vs. Final Position

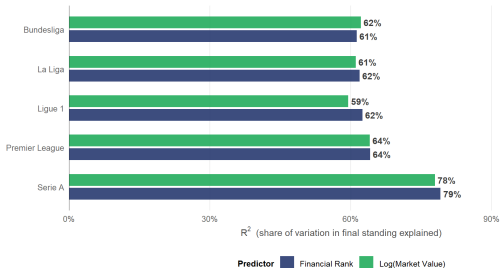
$$N = 975, p < 0.001$$

- **Gold diamonds** = five notable clubs whose finish surprised the model.
- Every league, every season: the upward trend holds.
- **Exceptions exist** — and they tell the real story.

The Effect Holds in Every League

Money Explains 61–79% of Final League Position

R^2 per league • Both predictors significant ($p < 0.001$) in all leagues



OLS: League_Position ~ predictor, estimated separately per league. Serie A shows the strongest relationship ($R^2 = 0.79$).

League

R^2

Bundesliga 0.613

La Liga 0.619

Ligue 1 0.625

Premier League 0.641

Serie A 0.791

Financial Rank alone explains 61–79 % of where a club finishes. In Serie A the figure rises to 79 %.

Model Results

Model H: The Best-Fitting Specification

Mdl	Formula	R^2	AIC
A	FR only	0.660	5099.8
B	FR + League FE	0.660	5107.4
C	Log(MV) only	0.533	5410.6
D	Log(MV) + League + Season	0.641	5164.3
H	FR + FR²	0.671	5070.5

OLS. Response: League position (1 = champion).

$N = 975$. Bold = best by AIC.

Model H: Estimated Equation

$$\widehat{\text{Pos}} = 0.42 + 1.23 \text{FR} - 0.02 \text{FR}^2$$

- Each rank step $\approx +1.2$ positions.
- Negative quadratic: the penalty for being poorest is smaller than the reward for being richest.
- Quadratic term: $F = 31.7$, $p < 0.001$.

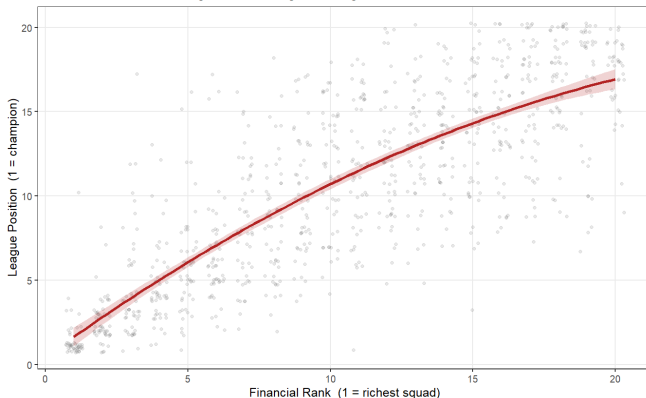
Statistical view: $\hat{\beta}_{\text{FR}} = 1.23$ (SE 0.08);
 $\hat{\beta}_{\text{FR}^2} = -0.020$ (SE 0.004).

Plain English: Being the richest club gives a big head start, but being the very poorest is less catastrophic than a simple linear rule

Diminishing Returns – Top Teams Compress at the Summit

Quadratic Fit: League Position vs Financial Rank (Model H)

Tests whether the financial advantage shows diminishing returns at high ranks



What the curve says

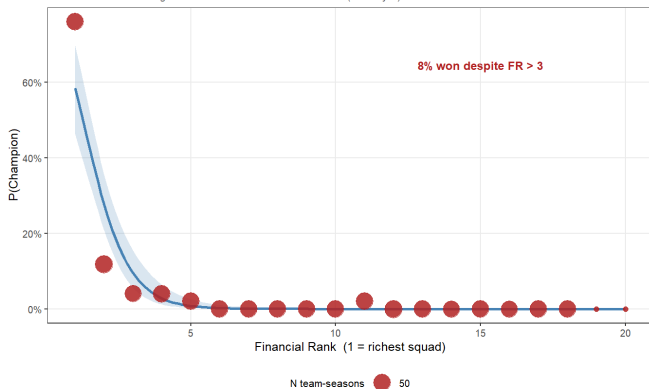
- FR 1 → 2: predicted finish moves from **1.6** to **2.8** ($\Delta = 1.2$ places).
- FR 15 → 16: penalty shrinks to ≈ 0.6 places.
- The curve flattens at the bottom: mid-table spending differences matter less.

A marginal euro spent by a top-2 club buys more league position than the same euro spent by a mid-table club.

Champion Probability Drops Sharply With Financial Rank

Predicted Probability of Championship vs Financial Rank (Model E)

Blue band = 95% CI of logistic fit • Red dots = observed win rate (sized by N)



FR	P(champion)
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1	58.5 %
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2	27.8 %
---	--------

3	9.5 %
---	-------

5	0.8 %
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11	<0.01 %
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Logit Model E:

Champion $\sim$ FR. McFadden $R^2 = 0.55$.

$\hat{\beta}_{FR} = -1.30$, $p < 0.001$.

Statistical view: Log-odds fall by 1.30 per rank step ($\approx 73\%$ odds reduction).

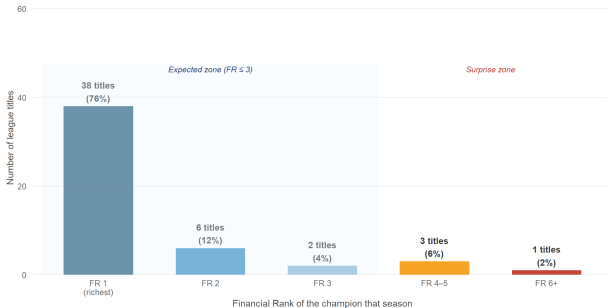
Plain English: Winning from outside the richest two clubs is statistically almost impossible —

The Exceptions

76 % of Titles Were Won by the League's Richest Club

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50 champions over 10 seasons across 5 leagues • 92% came from the financial top 3



FR 6+ = 4 champions: Leicester City (FR 11, 2015-16), Napoli (FR 5, 2024-25), Lille (FR 4, 2020-21), Liverpool (FR 4, 2024-25).

FR	Titles	%
1 (richest)	38	76 %
2	6	12 %
3	2	4 %
4-5	3	6 %
6+	1	2 %
Total	50	100 %

92 % of champions came from the financial top 3. Only 4 clubs (8 %) won from outside the richest three squads.

The Four Statistical Upsets – Ten Years of Data

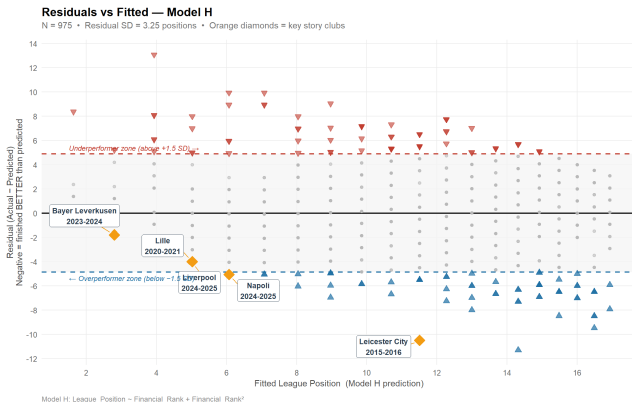
Club	Season	League	FR	Predicted	Residual
Leicester City	2015–16	Premier League	11	11.5	–10.5
Napoli	2024–25	Serie A	5	6.1	–5.1
Lille	2020–21	Ligue 1	4	5.0	–4.0
Liverpool	2024–25	Premier League	4	5.0	–4.0

A negative residual = the club finished better than its financial rank predicted. Leicester's –10.5 positions is the largest overperformance in the entire dataset.

Also noteworthy

Bayer Leverkusen 2023–24 (FR = 2): ended Bayern's 11-year Bundesliga streak. Residual = –1.8 (within normal range statistically, but historically significant).

Leicester City 2015–16 – A 10-Position Statistical Shock

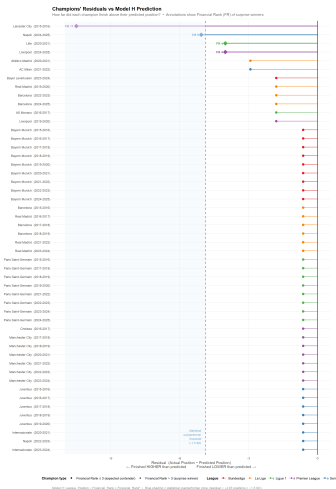


Residual = -10.5

Model predicted 11th place. They finished 1st.

- Only champion to exceed the **-1.5 SD threshold** in 10 years of data.
- FR = 11: squad valued at £161 M vs. league average £458 M.
- Coach: Claudio Ranieri.
Bookmaker odds: 5000-to-1.

How the Dotplot Tells the Full Story



Reading the chart

- Each dot = one league title.
- **Left of zero** = finished better than predicted.
- **Diamond** = $FR > 3$ (surprise winner); label shows FR.
- Blue shading = statistical overperformer zone ($< -1.5 SD$).

All 50 champions have negative residuals: every title

Robustness Checks

Three Tests – The Core Finding Holds

Test	Statistic	p -value	Verdict
Quadratic vs. linear	$F = 31.7$ (1 df)	< 0.001	Quadratic confirmed
League interaction (FR $\times$ League)	$F = 1.17$ (4 df)	0.32	Effect uniform
Time trend (FR $\times$ Season)	$F = 0.44$ (1 df)	0.51	Stable 2015–25
Heteroskedasticity	BP $\chi^2 = 45.5$	< 0.001	Present; HC1 SEs used
FR coeff. under HC1 robust SEs		< 0.001	Robust

The FR effect is statistically identical across all five leagues and has not strengthened or weakened over the ten seasons studied.

Statistical view: Breusch–Pagan $\chi^2 = 45.5$, $p < 0.001$; corrected with HC1 heteroskedasticity-robust SEs.

Plain English: Prediction errors vary in size across the position range, so we use corrected

Conclusions

Key Findings – Three Takeaways

1. Money predicts position strongly.

Financial Rank alone explains **67 %** of variation in final standing (Model H, $R^2 = 0.671$; $r = 0.813$ across 975 observations).

2. Surprises are statistically rare.

Only 4 of 50 champions (8 %) came from outside the richest three squads. Leicester City 2015–16 is the lone statistical outlier (residual = -10.5 positions).

3. The effect is stable and non-linear.

The quadratic term ($p < 0.001$) shows diminishing returns at the bottom; the time-trend interaction is not significant ($p = 0.51$).

For statisticians: Model H ($\hat{\beta}_{FR} = 1.23$, $\hat{\beta}_{FR^2} = -0.020$, $AIC = 5070.5$) outperforms all tested specifications.

For fans: being the richest club gives a 58 % championship probability. Being the 5th-richest drops that to under 1 %. Football still surprises us — just not very often.

Limitations

- Market value as proxy for quality (i.e. β_{MV})

Thank You – Questions?

Money and Success in Football

Does Financial Strength Predict League Position?

$$r = 0.813 \quad R^2 = 0.671 \quad N = 975$$

Five leagues · ten seasons · one answer: **yes, mostly**

Data: ESPN Standings + Transfermarkt

Period: 2015–16 to 2024–25

Models: OLS (A–H), Logit (E, F, J), Ordinal Logit (K)

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Appendix / Backup Slides Available

- Full model coefficient table (Models A–H)
- Cook's distance diagnostics
- Ordinal logistic results (Model K)
- Season-by-season champion FR table
- League interaction slopes (Model G)

The one exception everyone remembers: Leicester City 2015–16. FR = 11. Residual = –10.5 positions. The model said 11th. They won the league.

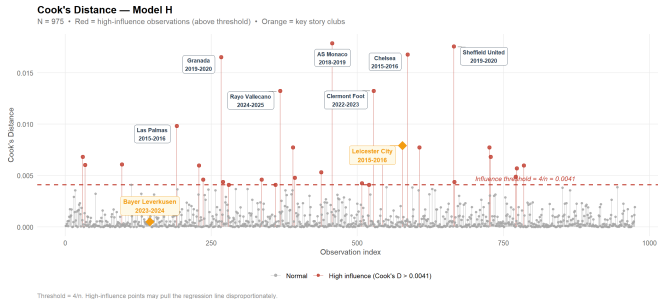
Appendix A – Full Model Comparison

Mdl	Formula	R^2	Adj. R^2	AIC	N
A	FR	0.660	0.660	5099.8	975
B	FR + League	0.660	0.659	5107.4	975
C	Log(MV)	0.533	0.532	5410.6	975
D	Log(MV) + League + Season	0.641	0.639	5164.3	975
G	FR $\times$ League	0.651	0.645	5110.7	975
H	FR + FR ²	0.671	0.670	5070.5	975
I	FR $\times$ Season	0.651	0.649	5103.4	975
E	Champion $\sim$ FR (Logit)	McFadden = 0.548		182.4	975
F	Champion $\sim$ Log(MV)+League (Logit)	McFadden = 0.592		173.1	975

OLS response: League_Position

(1 = champion). Bold = best OLS by AIC. Model F = best logistic.

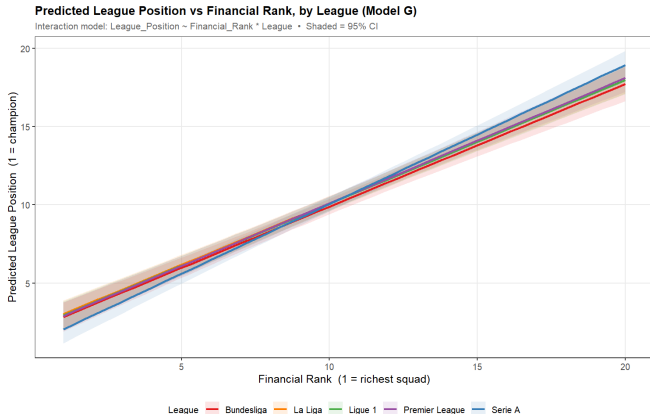
Appendix B – Cook's Distance Diagnostics



Interpreting Cook's D

- Threshold = $4/n = 0.0041$.
- High-influence points can pull the regression line disproportionately.
- Leicester City is both the **largest residual** and the **highest Cook's D**.
- Removing Leicester raises the intercept slightly but does not change slope direction.

Appendix C – Interaction Slopes by League (Model G)



Model G: $FR \times League$

- Interaction $F = 1.17$,
 $p = 0.32 \Rightarrow$ **not significant**.
- Slopes are visually parallel:
the financial advantage is the
same size in all five leagues.
- Serie A slope slightly steeper,
consistent with its highest R^2 .

**Money works the same way
in Munich, Madrid, Milan,
Marseille, and Manchester.
The mechanism is league-
invariant**